UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Sixth Semester – Fall 2022

Paper: Organic Chemistry Course Code: CHEM-317 [redacted]

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions. (6x5=30)

i. Give possible electronic transitions in S
a) methyl Benzoate b) Propanaldehyde

ii. Draw a stepwise mechanism for the following reaction.

$$CH_3COCl \xrightarrow[(2)\ H_2O]{(1)\ LiAlH_4} CH_3CH_2OH$$

iii. Explain why ethylene does not show C=C streching band while propylene does. S
List the types of vibrations which occur in CH_2=CH-CH_2-Br? S

iv. What is the term "persistance"? What are the radical scavengers? FR

v. Describe the effect of intermolecular and intramolecular hydrogen bonding on the position of absorption frequency of a compound? Explain with an example? S

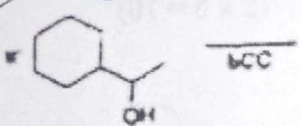
vi. Draw the organic products formed when 4-octyne is treated with each reagent.
a. H_2 (excess) + Pd-C b. H_2 + Lindlar catalyst c. Na, NH_3 d. [1] O_3; [2] H_2O

Answer the following questions.

Q. No. 2 i. Define the term chromophore & auxochrome with examples? How will you detect the presence of carbonyl group in aldehydes and ketones? (3) S (5)

ii. Write a note on applications of free radicals. (5) (2) FR

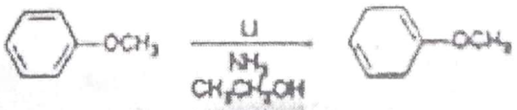
Q.No. 3 i. Complete the reactions. (5)

a. 

b. $CH_3CH_2CH_2CH_2OH \xrightarrow{PCC}$

c. 

ii. The Birch reduction is a dissolving metal reaction that converts substituted benzenes to 1,4-cyclohexadienes using Li and liquid ammonia in the presence of an alcohol. Draw a stepwise mechanism for the following Birch reduction. (5)



Q.No. 4 i. How would each of the following pairs of compounds differ in their IR spectra? (5) S

a.  and $HC{\equiv}CCH_2CH_2CH_3$

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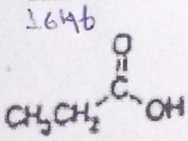
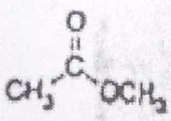
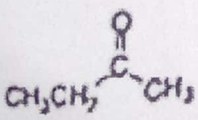
b.  and 

c.  and $CH_3CH{=}CHCH_2OH$

d. $HC{\equiv}CCH_2N(CH_2CH_3)_2$ and $CH_3(CH_2)_4C{\equiv}N$

e. $CH_3C{\equiv}CCH_3$ and $CH_3CH_2C{\equiv}CH$

ii. Write down various steps for the following transformations? (5)
Benzene to 2-naphthanoic acid Acetone to trimethyl acetic acid

Umer Abdullah GPGC

UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Sixth Semester – Spring 2022

Time: 3 Hrs.

Course Code: CHEM-317

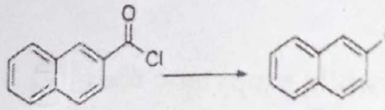
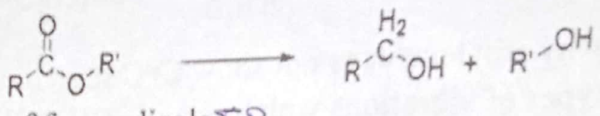
Paper: Organic Chemistry

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVI

Q.1. Answer the following short questions. (6x5=30)

i. Explain Oppeneaur oxidation give mechanism and describe its role in oxidation of secondary alcohols?

ii. How would you carry out following conversions? Give name and mechanism?

iii. Describe Perch method for detection of free radicals. FR

iv. Describe instrumentation and sample handling in UV/VIS spectrophotometer. S

v. Describe Clemmensen and Wolf-Kishner reduction with mechanisam for aldehydes and ketones.

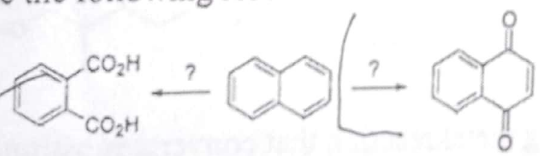
vi. Define lambert beer law and Differentiate hypochromic and bathochromic shift? S

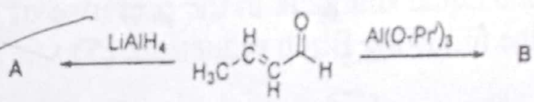
Answer the following questions.

Q. No. 2 i. Give mechanism of Corey-Kim oxidation of primary alcohols to aldehyde. (5)

ii. What is reductive amination? Give different variations and modifications of reaction. (5)

Q. No. 3 Complete the following reactions and draw their mechanisms: (2 x 5 = 10)

a) 

b) 

Q. No. 4. i. Write a note on applications of infrared spectroscopy. S (5)

ii. Explain the method for the selective oxidation of $ArCH_3$. (5)

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UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Sixth Semester – 2020

Roll No. ....

Paper: Organic Chemistry
Course Code: CHEM-317 Part – II

Time: 2 Hrs. 45 Min. Marks: 50

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q.2. Answer the following questions. (5x4=20)

I) Explain Swern oxidation, give mechanism and example for oxidation of primary alcohols.

II) Write a short note on Birch reduction of benzaldehyde and methoxybenzene.

III) Describe two synthetic applications of free radicals. FR

IV) Describe types of vibration in IR spectroscopy. S

V) Give four reactions with reagents to reduce aldehydes or ketones to hydrocarbons.

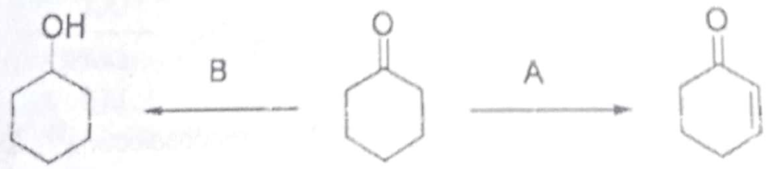
Q. No. 3.

I) Write a short note on ozonolysis of alkenes with mechanism and examples. [5]

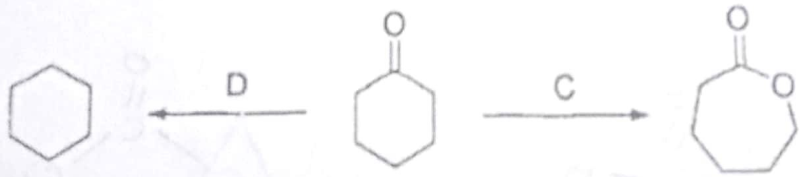
II) Describe hydroboration of alkenes to yield alkanes and alcohols. [5]

Q. No. 4. Complete the following reactions and draw their mechanisms. [2 × 5 = 10]

i



ii



Q. No. 5. A) Write a note on applications of UV/Vis. spectroscopy. S [5]

B) Give different methods for the generation of free radicals. FR [5]

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UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Sixth Semester – 2019

Roll No. ....

Paper: Organic Chemistry
Course Code: CHEM-317 Part – II

Time: 2 Hrs. 45 Min. Marks: 50

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q.2. Answer the following questions. **(4x5=20)**

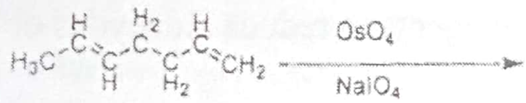
i) Give possible electronic transitions in Ethyl benzoate and Acetaldehyde S

ii) Discuss Birch reduction with suitable example and mechanism.

iii) Explain why ethylene does not show C=C stretching band while propylene does. S

iv) What is role of free radicals in aging? FR

v) Write product and mechanism of given reaction



Q.3

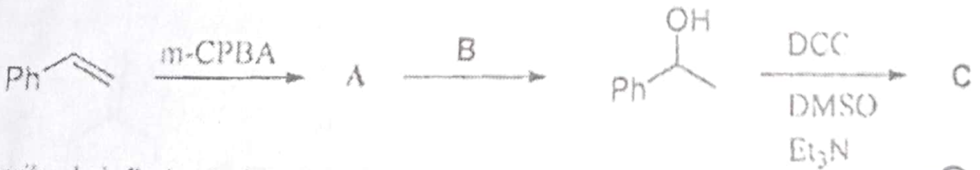
i) What a note on Swern oxidation of alcohols? (5)

ii) Discuss different methods for oxidative cleavage of double bond? (5)

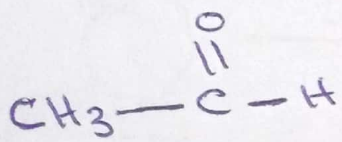
Q.4. Describe sample handling in infrared spectroscopic technique? S (10)

Q.5.

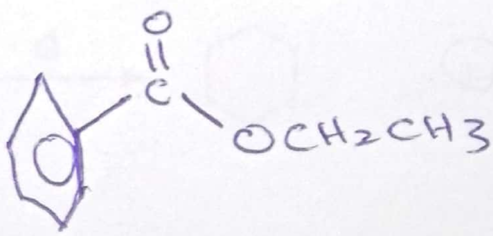
a) How would you complete these reactions with mechanism? (6)



b) Describe briefly hydroboration with mechanism and regeoselectivity? FR (4)



Acetaldehyde



Ethyl benzoate

UNIVERSITY OF THE PUNJAB

Sixth Semester - 2018

Examination: B.S. 4 Years Programme

Roll No. ..0S4018....

PAPER: Organic Chemistry
Course Code: CHEM-317 Part – II

TIME ALLOWED: 2 Hrs. & 45 Mints.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

Q. No. 2. Give the short answers of the following questions. (4 × 5 = 20)

I. What is Beers-Lambert law? S
II. What is the effect of ring size on IR absorptions in cyclic ketones? S
III. Describe Clemmensen reduction with example and mechanism?
IV. Describe epoxidation of alkenes with example and mechanism?
V. Describe two factors affecting the stability of free radicals? FR

Q. No. 3. Describe two different methods for the following conversions with mechanism?

(i) Alkene to *Cis*-diol (ii) Acid chloride to aldehyde (5+5=10)

Q. No. 4. Write a note on the followings with example. S

a)- Infrared Spectrophotometer (Instrumentation) (5)

b)- Hypochromic effect and Hyperchromic Effect (5)

Q. No. 5. Write a note on the followings?

a)- Applications of UV/Vis. Spectroscopy? S (5)

b)- Applications of free radical reactions? FR (5)

UNIVERSITY OF THE PUNJAB

Sixth Semester - 2017

Examination: B.S. 4 Years Programme

Roll No. ....

PAPER: Organic Chemistry
Course Code: CHEM-317

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

Q. NO. 2. Give the short answer of the following questions. (4 × 5 = 20)

i. Name different types of UV/Vis. transitions possible in Benzamide and Ethyl benzoate? S
ii. Why conjugation of C=O with C=C lowers its stretching frequency? S
iii. Describe Birch reduction with example and mechanism?
iv. Write a brief note on Wolf-Kishner reduction with example and mechanism.
v. Describe Oppenauer oxidation of secondary alcohols. Give example and mechanism.

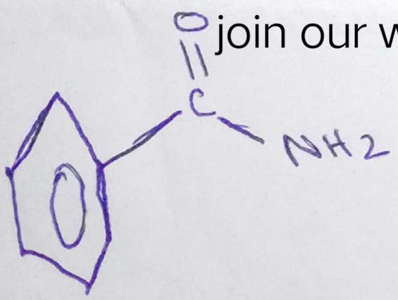
Q. NO. 3. Describe the following reactions with suitable examples and mechanism. (10)

(i) Ozonolysis of Alkenes (ii) Hydroboration of Alkenes

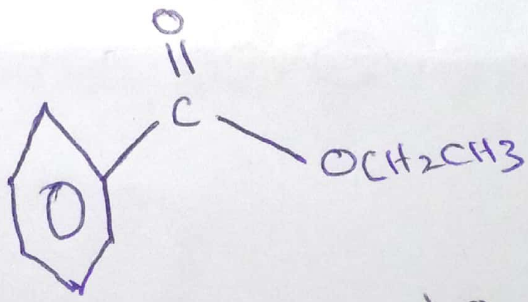
Q. NO. 4. Discuss in detail the various types of vibrations and factors influencing these vibrations in IR Spectroscopy. S (10)

Q, NO. 5. Write a detailed note on the followings? (10)

(i) Applications of UV/Vis. Spectroscopy in Chemistry S
(ii) Factors affecting stability of free radicals FR



Benzamide



Ethyl benzoate

UNIVERSITY OF THE PUNJAB

Sixth Semester 2015
Examination: B.S. 4 Years Programme

Roll No. ....

Paper: Organic Chemistry
Paper Code: CHEM-317

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

Subjective Marks: 50

Q.2 Answer the following questions briefly, all questions carry equal marks. (10 x 2 = 20)

I. What is Baeyer-Villiger Oxidation?

II. Describe Mozingo method of reduction for ketones.

III. What is Captodative Effect? FR

IV. What is mean by finger print region? S

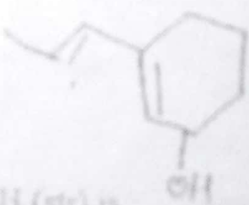
V. What is the difference between Wolff-Kishner reduction and Clemmensen reduction?

VI. The energy difference between the ground state and excited state of an atom is 4.4×10^{-19} J. Calculate the λ required to produce this transition. S

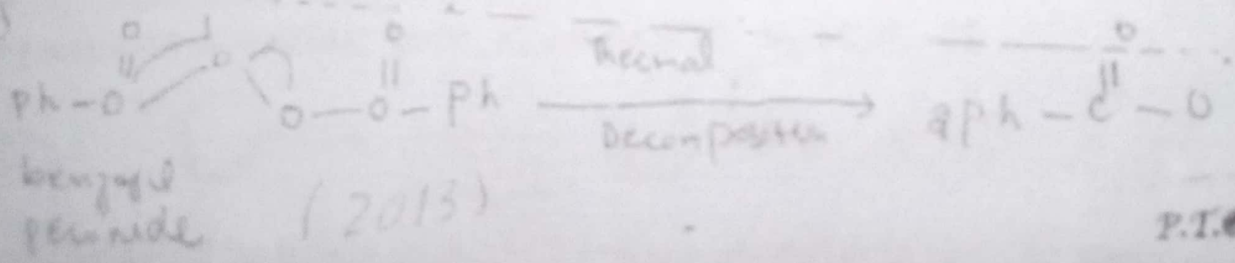
VII. Define the term "persistent." What are radical scavengers? FR

VIII. What product do you expect from thermal decomposition of benzoyl peroxide? FR

IX. Complete the following reaction.



X. When 10 ml of CCl_4 is added to 2 ml of C_2H_5OH then O-H (str) is changed. Why? S



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